Name: ____________________________ Date: ________________ Period: _________

Spiral Review — Key

1. A particle's x-position is $x(t) = t^2 - 6t$. Complete the table of values.

t	0	1	2	3	4	5	6
$x(t)$	0	-5	-8	-9	-8	-5	0

2. Find the vertex of $y = -t^2 + 4t + 5$. What is the maximum value of y?

Vertex at $t = -4/(2 \cdot (-1)) = 2$. Maximum value: $y(2) = -(4) + 8 + 5 = 9$.

Teacher Note

Students should use the vertex formula $t = -b/(2a)$ explicitly. Reinforce that the maximum *value* is $y(2) = 9$, not $t = 2$. This is the same technique they will use in the lesson to find vertical extrema of parametric motion.

3. Solve $t^2 - 9 = 0$.

$t = \pm 3$

A particle moves with $x(t) = t^2 - 9$. What do the solutions above tell you about the particle's position?

At $t = -3$ and $t = 3$, the particle has $x = 0$, so it is on the y-axis at those moments. These are y-intercepts of the parametric curve.